Andrew **Romans**

Carlsbad, CA

andrewromans@proton.me | andrewromans.com | aromans | andrew-romans

Summary

Security engineer and vulnerability researcher specializing in reverse engineering and malware analysis, with applied machine learning experience across both halves of the field. Three and a half years in offensive and defensive security, following five years of machine learning and software engineering. Most recently applied both in an M.S. capstone fine-tuning code LLMs to automate binary deobfuscation. Fluent in low-level systems — C, x86-64, ARM, and MIPS — and in the modern training stack, including PyTorch, HuggingFace, and PEFT. Competitive CTF player: solved every challenge in Flare-On 12 for a top 100 finish, and earned High Performer in the NSA Codebreaker Challenge in 2022 and 2025.

Work Experience

Viasat
Carlsbad, CA

Vulnerability Researcher & Security Engineer — *Jan 2023 - Present*

- Conduct in-depth malware analysis and reverse engineering of sophisticated threats, dissecting binaries to identify exploitation techniques, command-and-control mechanisms, and indicators of compromise.
- Perform vulnerability research and penetration testing across network infrastructure, embedded systems, and applications.
- Leverage low-level debugging, disassembly (IDA Pro, Ghidra, Binary Ninja), and dynamic analysis and emulation tooling (AFL++, Unicorn, angr) for vulnerability research and malware analysis.
- *Much of this work is non-public; happy to discuss scope and methodology directly.*

Air Force
Warner Robins, GA

Computer Scientist — *Feb 2021 - Jan 2023*

- Led artificial intelligence efforts for the 577th within the Distributed Common Ground System (DCGS), spanning natural language processing, computer vision, and regression modeling.
- Developed natural language processing models for a system designed to detect and remediate operational errors in real time.
- Served as an internal AI consultancy for the organization: development teams across the unit brought their machine learning work to us for guidance on approach, model selection, and evaluation design.
- Designed the team's CI/CD system architecture and drove secure development practices throughout the software lifecycle.

Shadow Health
Gainesville, FL

Game Developer — *Feb 2018 - Feb 2021*

- Orchestrated the development of multiple high-fidelity 3D simulations for nursing colleges in over 1,800 programs worldwide.

Projects

Perseus
Perseus

Automated Malware Deobfuscation via LLM Fine-Tuning — M.S. Capstone, Georgia Tech — *May 2026*

- Built an end-to-end pipeline that generates verifiable deobfuscation tasks with programmatic ground truth using Tigress obfuscation (mixed boolean arithmetic, control flow flattening, virtualization) over the 1M-program AnghaBench corpus, compiled to x86-64 ELF, disassembled per-function against the .symtab and .dynsym symbol tables, and paired with clean-assembly targets.
- Fine-tuned four code LLMs with QLoRA on a single H100.
- Validated against a Flare-On 12 challenge binary - fine-tuned models collapsed an 18-instruction obfuscated sequence from a Windows PE down to 5 correct instructions, on an unseen binary, executable format, and obfuscator, where the base models failed to produce assembly at all.
- Diagnosed the virtualization failure mode as a context-length limit that drove the model to a degenerate fixed-length output regardless of input, and identified control-flow-graph-conditioned input as the fix, a direction confirmed by independently published work.
- Designed the evaluation, including a normalization pass for position-dependent operands, after finding that the surface metric was penalizing semantically correct output.

BSidesSD 2026 Hardware Hacking CTF
Repository

Design and firmware — *Apr 2026*

- Created an introductory hardware hacking CTF on an rp2040.
- Taught UART/SPI protocols, logic analyzer squashfs extraction, EMFI glitching, and more.

DEF CON Badge
Independent

Hardware design and firmware (In progress for DC 35) — *Aug 2026 - Present*

- Designing and building an electronic badge with embedded CTF challenges.

CTF & Recognition

2025	**Flare-On 12**, Solved all challenges; top 100 finish	*Mandiant*
2025	**NSA Codebreaker Challenge**, High Performer, solved all but one task	*NSA*
2022	**NSA Codebreaker Challenge**, High Performer, solved all but one task; published full writeups	*NSA*

Technical Skills

Reverse Engineering	IDA Pro, Ghidra, Binary Ninja, GDB, x86-64 / ARM / MIPS, ELF and PE internals
Vulnerability Research	AFL++, Unicorn, angr, fuzzing harness design, symbolic execution, penetration testing, malware analysis
Machine Learning	PyTorch, HuggingFace Transformers, PEFT (LoRA / QLoRA), supervised fine-tuning, evaluation design
Languages & Infrastructure	C, C++, Python, x86-64 assembly, CI/CD architecture, Docker, single-GPU and cloud GPU training

Education

Georgia Institute of Technology *Atlanta, GA*

M.S. in Cybersecurity *Jan 2019 - May 2026*

- Finished all but two courses toward an M.S. in Computer Science, AI specialization, before pivoting to Cybersecurity. Capstone: Perseus, automated malware deobfuscation via LLM fine-tuning.

Kennesaw State University *Kennesaw, GA*

B.S. in Computer Game Design and Development *Aug 2013 - May 2017*

Security Clearance

TS/SCI *Feb 2021 - Present*